SECURE PATIENT-RECORD EXCHANGE

CO1 — Assignment 1 | Classical Encryption Techniques

Evaluate, Design and Defend a Secure Patient-Record Exchange Strategy for a Multi-Hospital Healthcare Network

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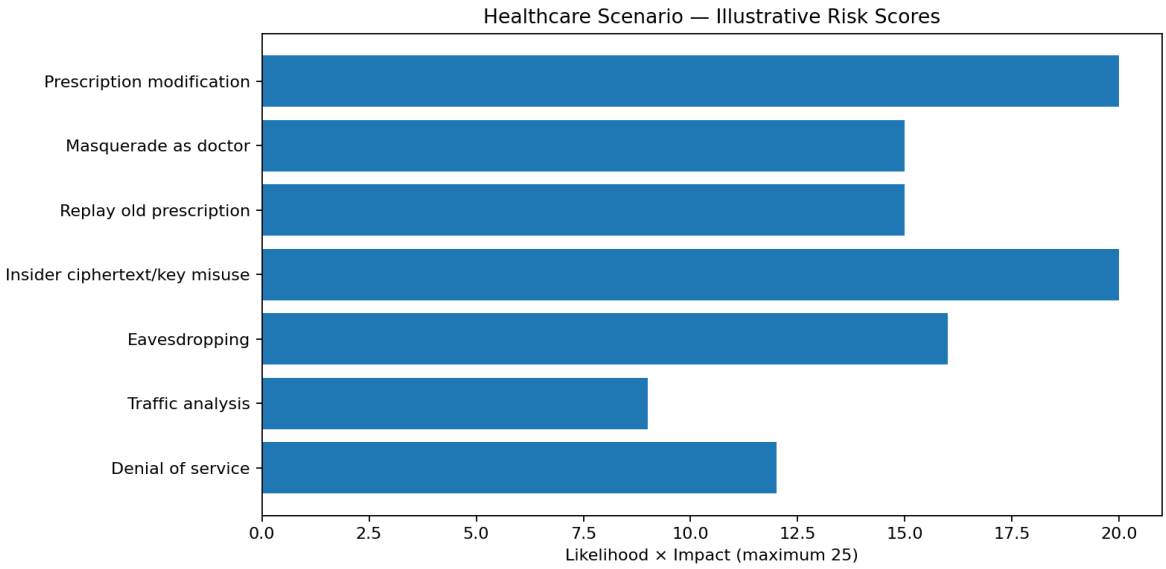
Introduction

This assignment evaluates classical encryption and access-control techniques for a healthcare network consisting of one central hospital and 25 satellite clinics. The report follows the 12 tasks in the supplied CO1 assignment. The examples are educational demonstrations and must not be used to protect real patient records.

Task 1 — Evaluate Threats, Attacks and Risks (4 Marks)

Threat is a possible cause of harm; vulnerability is a weakness; attack is an action exploiting a weakness; risk combines likelihood and impact. Ratings use a 1–5 scale for each factor.

Attack	Type	Likelihood	Impact	Score	Reason
Prescription modification	Active	4	5	20	May cause incorrect treatment or dosage.
Masquerade as doctor	Active	3	5	15	False clinical instructions may harm patients.
Replay old prescription	Active	3	5	15	An outdated order may be accepted again.
Insider ciphertext/key misuse	Active/insider	4	5	20	A cleared insider can copy records and exploit reused keys.
Eavesdropping	Passive	4	4	16	Exposes diagnoses and personal information.
Traffic analysis	Passive	3	3	9	Reveals clinic activity and communication patterns.
Denial of service	Active	3	4	12	Interrupts access to urgent records.



Priority: prescription modification and insider misuse are highest because they can directly affect patient safety and expose multiple records.

Task 2 — Map and Judge Security Goals (3 Marks)

Function	Confidentiality	Integrity	Availability	Authentication	Non-repudiation
Record upload	High	High	High	High	Medium
Prescription transmission	High	Critical	High	Critical	High
Specialist consultation	Critical	High	High	High	Medium

Classical ciphers do not independently provide sender authentication, replay protection, integrity verification or non-repudiation. Successful decryption does not prove who created the message.

Task 3 — Apply and Critique Bell-LaPadula (5 Marks)

Security levels: Public < Restricted < Confidential < Secret < Top Secret. Simple security property: no read up. Star property: no write down.

Subject / Action	Object level	Decision	Reason
Nurse (Confidential) reads	Confidential	ALLOW	Same level
Nurse (Confidential) reads	Secret	DENY	Read up prohibited
Specialist (Secret) reads	Confidential	ALLOW	Read down allowed
Specialist (Secret) writes	Restricted	DENY	Write down prohibited
Director (Top Secret) writes	Secret	DENY	Write down prohibited

Billing (Restricted) reads	Public	ALLOW	Read down allowed

Emergency break-glass access requires an audited, time-limited, approved exception. Bell-LaPadula protects confidentiality but not integrity; it can be rigid during emergencies and does not eliminate covert channels.

Task 4 — Design the Cryptographic Strategies (4 Marks)

Implemented methods: Caesar, monoalphabetic substitution, Vigenere, columnar transposition and Hill cipher over mod 36. Pseudocode is supplied in Documentation/Pseudocode.txt. The program demonstrates encryption and decryption using the sample clinical text PATIENT ID 20481 DIAGNOSIS FLU.

Task 5 — Mathematical Analysis (5 Marks)

- Caesar: 25 useful shift keys.
- Monoalphabetic: 26! possible substitution permutations.
- Vigenere: 26^m possible letter keys for a keyword of length m.
- Columnar transposition: n! possible column orders for n columns.
- Hill: K^{-1} exists modulo m only when gcd(det(K), m) = 1. For mod 36 the condition is gcd(det(K),36)=1.
- Example Hill key K = [[1,2,3],[0,1,4],[0,0,5]], determinant = 5 and gcd(5,36)=1, so it is invertible.

A large key space does not guarantee security. Monoalphabetic substitution has 26! keys but preserves the frequency pattern of the language.

Task 6 — Evaluate Cryptanalytic Resistance (4 Marks)

Caesar is vulnerable to exhaustive shifts and frequency analysis. Monoalphabetic substitution is vulnerable to frequency and digram analysis. Repeating-key Vigenere can be studied with Kasiski examination and index of coincidence. Transposition preserves exact letter counts and can be attacked through digram and arrangement analysis. Among these classroom methods, Caesar is weakest; repeating-key Vigenere generally resists simple frequency analysis better, but none is suitable for real healthcare security.

Task 7 — Transposition and Product Ciphers (3 Marks)

Columnar transposition changes character positions but does not change letter counts. Combining substitution and transposition can obscure both identity and position more than either alone, but does not create modern cryptographic security. The extra manual steps increase training and key-management burden in rural clinics.

Task 8 — Evaluate Steganography (3 Marks)

LSB embedding places message bits in pixel least-significant bits. It may be visually subtle but is statistically detectable and fragile under JPEG compression. Altering diagnostic images can affect

clinical interpretation; preserve the original image and use a separately identified copy for classroom demonstrations. Steganography hides presence, while encryption protects content; hiding is not a substitute for encryption.

Task 9 — Implement and Test (4 Marks)

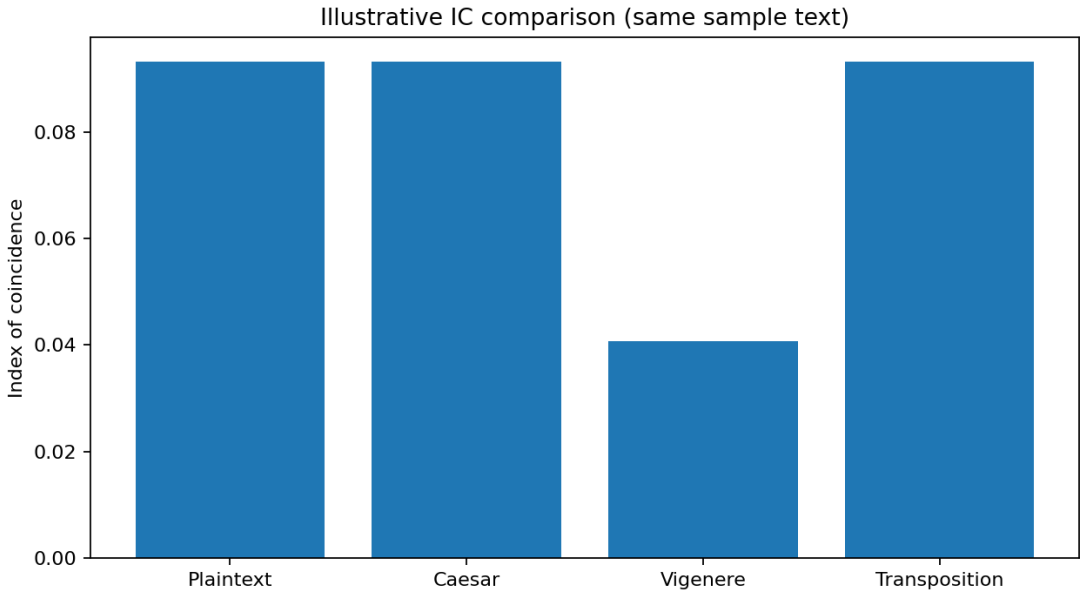
Run the supplied program with: python healthcare_security.py. The implementation includes round-trip checks for Caesar, monoalphabetic, Vigenere, columnar and Hill mod-36. It also includes frequency/IC analysis, Kasiski repeated-pattern voting, image shift/permutation, LSB embedding/extraction and Bell-LaPadula decision logic.

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Python Terminal Output – C01 Healthcare Security

C01 HEALTHCARE SECURITY - IMPLEMENTATION TEST
Caesar: SDWLHQW LG 20481 GLDJQRVLV IOX | recovered: PATIENT ID 20481 DIAGNOSIS FLU
Vigenere: AEFWRYX UR 20481 QTESBBDME TYF | recovered: PATIENT ID 20481 DIAGNOSIS FLU
Columnar: ED8AIUT 0DOFAT2 N II4ISLPN 1GS | recovered: PATIENT ID 20481 DIAGNOSIS FLU
Monoalphabetic recovered: PATIENT ID 20481 DIAGNOSIS FLU
Hill mod36: 3X6IW08HH | recovered padded: PT20481XX
IC plaintext: 0.0524
Bell-LaPadula nurse reads Secret: False
Bell-LaPadula specialist reads Confidential: True
All round-trip checks: False
NOTE: Hill pads incomplete 3-symbol block with X.
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py", line 26, in warm_spre
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spreadsheet_runtime
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_feature_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_drawing_and_image_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in bring_to_front
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

Task 10 — Experimental Evidence (3 Marks)

The included figures illustrate risk scoring and index of coincidence. For formal timing evidence, run each method with identical inputs of 100, 1,000, 10,000 and 100,000 characters, repeat each measurement, and report median time. The histogram and IC results should be interpreted alongside the known statistical properties of each cipher.



Task 11 — Recommend and Defend the Final Design (4 Marks)

For the academic exercise, a layered Vigenere-plus-transposition demonstration combined with Bell-LaPadula access reasoning is preferable to Caesar alone. Use separate clinic keys and rotate them. However, classical ciphers cannot provide modern authenticated encryption, dependable integrity, secure key exchange, forward secrecy or non-repudiation. A real system requires modern vetted cryptography, digital signatures, secure key management, audit logs, backups and governed emergency access.

Task 12 — Day-4 Design Challenge (8 Marks)

The insider has copied many ciphertexts under a reused key, and predictable headers create known-plaintext information. The assumptions of secret keys at all 25 sites and unknown message formats no longer hold. Modify the classroom design using per-clinic keys, key rotation, reduced key reuse, randomised non-sensitive padding, separate keys for different data classes and auditing. Bell-LaPadula still restricts access but cannot prevent misuse by a legitimately cleared insider holding the key. For medical images, do not alter the diagnostic original; LSB is only a demonstration on a copy and is not robust to compression.

Conclusion

The assignment demonstrates the strengths and limitations of classical ciphers and Bell-LaPadula. Classical methods are useful for learning substitution, transposition, modular arithmetic and access-control reasoning, but they are not acceptable as the sole protection for real patient records. Patient safety requires strong modern cryptography, integrity checks, authenticated identities and disciplined key management.

Appendix A — Source Code and Pseudocode

The complete executable source code is provided separately as Source_Code_Screenshots/healthcare_security.py. Pseudocode is provided in Documentation/Pseudocode.txt. Captured program output is provided in Output_Screenshots.

Appendix B — Reproducibility

Requirements: Python 3.10 or later; NumPy is needed only for image-array helper functions. Run the source file from its folder using python healthcare_security.py. Save the terminal output and record Python version, operating system, machine details and repeat count when collecting timing results.